NOETICA MASTER BLUEPRINT

PART 03

Discovery Framework

Executive Summary

The Discovery Framework is the core intellectual foundation of Noetica.

Most scientific systems treat papers as the primary object.

Noetica treats discoveries as the primary object.

A paper is merely evidence that a discovery exists.

The discovery itself is the entity that matters.

The Fundamental Principle

Traditional Systems:

Paper ↓ Citation ↓ Ranking

Noetica:

Discovery ↓ Evidence ↓ Significance ↓ Knowledge Connections ↓ Civilizational Impact

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What Is A Discovery?

A discovery is any new insight, principle, technology, method, theory, dataset, capability, or understanding that advances human knowledge.

A discovery may be represented by:

- One paper
- Multiple papers
- Patents
- Grants
- Datasets
- Open-source projects

- Products
- Conferences
- Scientific communities

Discovery Categories

Scientific Discoveries

Examples:

- DNA Structure
- Higgs Boson
- Gravitational Waves
- CRISPR

Mathematical Discoveries

Examples:

- Calculus
- Information Theory
- Gödel's Incompleteness Theorems
- Category Theory

Technological Discoveries

Examples:

- Transistor
- Internet
- GPS
- AlphaFold

Methodological Discoveries

Examples:

- PCR
- BLAST
- Deep Learning
- Transformer Architecture

Dataset Discoveries

Examples:

- Human Genome
- ImageNet
- AlphaFold Database

Interdisciplinary Discoveries

Examples:

- AI Drug Discovery
- Quantum Biology
- Computational Social Science

Discovery Cluster Model

Noetica never treats papers as isolated objects.

Instead:

One Discovery

↓

Many Supporting Assets

Example:

CRISPR Discovery Cluster

Contains:

- Research Papers
- Patents
- Grants
- Clinical Trials
- Startups
- Products
- Datasets
- Conferences

Everything is grouped around the discovery.

Discovery Object Schema

Every discovery receives a unique identity.

Core Fields:

Discovery ID

Discovery Name

Discovery Type

Discovery Description

Date First Observed

Date Confirmed

Status

Lifecycle Stage

Primary Fields

Related Fields

Evidence Sources

Significance Scores

Knowledge Connections

Forecast Information

Discovery Lifecycle Engine

Every discovery progresses through stages.

Stage 0

Speculative

Characteristics:

- Early ideas

- Hypotheses
- Weak evidence

Examples:

- Room-temperature superconductivity claims
- Early quantum gravity proposals

Stage 1

Emerging

Characteristics:

- Initial evidence
- Limited adoption
- Growing attention

Stage 2

Growing

Characteristics:

- More publications
- More researchers
- Independent validation begins

Stage 3

Breakthrough

Characteristics:

- Strong evidence
- Major scientific attention
- Significant impact

Examples:

- AlphaFold
- CRISPR

Stage 4

Established

Characteristics:

- Accepted by the field
- Broad usage

Stage 5

Foundational

Characteristics:

- Enables future discoveries

Examples:

- PCR
- Information Theory

Stage 6

Civilizational

Characteristics:

- Changes humanity itself

Examples:

- Calculus
- Electricity
- Internet
- DNA

Stage 7

Declining

Characteristics:

- Replaced by better approaches

Stage 8

Historical

Characteristics:

- Primarily of historical interest

Discovery Relationships

Discoveries never exist in isolation.

Each discovery can have:

Enabled By

Influences

Competes With

Extends

Contradicts

Supports

Depends On

Replaced By

Derived From

Leads To

Example Discovery Graph

Boolean Logic

↓

Information Theory

↓

Computing

↓

Machine Learning

↓

Transformers

↓

Large Language Models

↓

AI Scientists

Discovery Momentum Engine

Purpose:

Measure acceleration.

Signals

Publication Growth

Citation Growth

Patent Growth

Funding Growth

Researcher Growth

Open Source Growth

Conference Growth

Dataset Adoption

Momentum Categories

Dormant

Slow Growth

Growing

Accelerating

Explosive

Transformational

Discovery Importance Levels

Every discovery receives importance classifications.

Local

Important within one subfield.

Field-Level

Changes an entire field.

Cross-Disciplinary

Changes multiple fields.

Scientific

Changes science broadly.

Civilizational

Changes humanity broadly.

Discovery Persistence

Many discoveries appear briefly and disappear.

Noetica tracks persistence.

Questions:

Does this discovery still matter after:

- 1 month?
- 1 year?
- 5 years?
- 20 years?

Persistence is a stronger signal than hype.

Discovery Lineage

Every discovery should have ancestry.

Example:

Mendel

↓

Genetics

↓

DNA

↓

Genome Sequencing

↓

Human Genome

↓

Bioinformatics

↓

CRISPR

↓

Synthetic Biology

This allows users to understand how discoveries evolve through time.

Discovery Descendants

Every discovery should also track descendants.

Example:

Transformer Architecture

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BERT

↓

GPT

↓

Protein Language Models

↓

AI Scientists

This allows future influence analysis.

Discovery Confidence Score

Measures confidence in the discovery itself.

Inputs:

Evidence Quality

Replication

Source Reliability

Consensus Strength

Methodological Quality

Discovery Maturity Score

Measures maturity.

Inputs:

Age

Adoption

Replication

Community Acceptance

Practical Use

Discovery Novelty Score

Measures originality.

Inputs:

Semantic Distance

Conceptual Innovation

Methodological Difference

Cross-Field Uniqueness

Discovery Registry

The permanent record of discoveries.

Every discovery ever tracked by Noetica enters the registry.

Nothing is deleted.

Status changes.

Importance changes.

Understanding evolves.

But the discovery remains part of the historical record.

Discovery Time Horizons

Noetica tracks discoveries across three horizons.

Foundational Timeline

5000+ years

Question:

What changed civilization?

Examples:

- Zero
- Calculus
- Evolution
- Relativity

Modern Timeline

50 years

Question:

What changed science?

Examples:

- PCR
- Human Genome
- CRISPR
- AlphaFold

Frontier Timeline

5 years

Question:

What may change the future?

Examples:

- AI Scientists
- Quantum Networks
- Synthetic Cells

Discovery Framework Principles

1. Discoveries are primary entities.

2. Papers are evidence.

3. Discoveries evolve over time.

4. Discoveries have ancestry and descendants.

5. Discoveries exist within knowledge networks.

6. Importance changes with time.

7. Persistence matters more than hype.

8. Civilizational impact matters more than popularity.

9. Discovery clusters are superior to isolated publications.

10. Understanding discovery evolution is more valuable than tracking papers.

End State

The Discovery Framework transforms Noetica from a paper-tracking platform into a discovery-tracking intelligence system capable of understanding how human knowledge emerges, evolves, connects, and shapes the future.